Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation of the **Smart Lender – AI Based Loan Approval Prediction** project. The application uses Machine Learning to predict whether a loan application is likely to be approved based on customer details. The solution demonstrates clean coding practices, proper project structure, and a functional web application developed using Flask.

Instructions:

1. The application is developed using **Python, Flask, HTML, CSS, NumPy, Pickle, and Machine Learning**.
2. The system predicts loan approval using a trained ML model based on applicant information.
3. The project includes source code, trained model (best_model.pkl), scaler (scaler.pkl), HTML templates, and required configuration files.
4. Attach your GitHub repository link below.

Solution Summary

Field	Details
Repository Link / URL	GitHub - shabeenashaik0506-droid/Smart-Lenders-Project · GitHub
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask
Key Features Implemented	• Loan Approval Prediction using Machine Learning • User-friendly Web Interface

	• Data Preprocessing using Standard Scaler • Real-time Prediction Result • Input Validation using Flask Forms
Pending / Incomplete Features	• User Login System • Database Integration • Loan History Storage • Email Notification
Setup / Run Instructions	1. Install required libraries (Flask, NumPy, Scikit-learn). 2. Keep best_model.pkl and scaler.pkl in the project folder. 3. Run python app.py. 4. Open http://127.0.0.1:5000 in the browser. 5. Enter applicant details and click **Predict** to view the loan approval result.

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: ☐ Machine Learning model is loaded using **Pickle**.
☐ Input data is normalized using **StandardScaler** before prediction.
☐ Flask routes (/, /input, /predict) manage navigation and prediction.
☐ Prediction result displays **Loan Approved** or **Loan Rejected** based on the trained model.